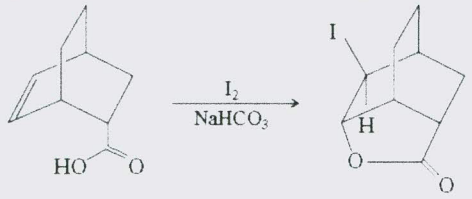
Organic Cume, October 14, 2011, Problem from Organic Chemistry, authors Carey and Giuliano, 8th Edition

6.55

A mixture of three alkenes (A, B, and C) was obtained by dehydration of 1,2-dimethyl cyclohexanol. The composition of the mixture was A (3%), B (31%), and C (66%). Catalytic hydrogenation of A, B, or C gave 1,2-dimethylcyclohexane. The three alkenes can be equilibrated by heating with sulfuric acid to give a mixture containing A (0%), B (15%), and C (85%). Identify A, B, and C.

6.64

The following reaction was performed as part of a research program sponsored by the National Institutes of Health to develop therapeutic agents for the treatment of cocaine addiction. Using what you have seen about the reactions of halogens with alkenes, propose a mechanism for this process



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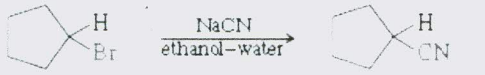
8.47

The reaction of cyclopentyl bromide with sodium cyanide to give cyclopentyl cyanide

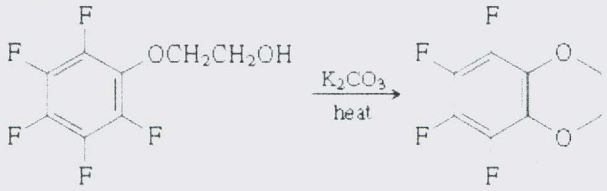


Cyclopentyl bromide Cyclopentyl cyanide

proceeds faster if a small amount of sodium iodide is added to the reaction mixture. Can you suggest a reasonable mechanism to explain the catalytic function of sodium iodide?

9.33

(*Z*)-9-Tricosene [(*Z*)-$CH_3(CH_2)_7CHPCH(CH_2)_{12}CH_3$] is the sex pheromone of the female housefl y. Synthetic (*Z*)-9-tricosene is used as bait to lure male fl ies to traps that contain insecticide. Using acetylene and alcohols of your choice as starting materials, along with any necessary inorganic reagents, show how you could prepare (*Z*)-9-tricosene.

12.72

Suggest a reasonable mechanism for the following reaction: